

Curriculum Vitae – Alessandra Mastrobuono-Battisti

PERSONAL INFORMATION

Family name, First name: Mastrobuono Battisti, Alessandra

Researcher unique identifier(s): ORCID: 0000-0002-2386-9142; SCOPUS ID: 35174814200

Nationality: Italian; **Family status:** Married, 2 children

Web site: <http://alessandramb.github.io/>

CURRENT POSITION

2024 – **Tenure Track Assistant Professor (RTDb)**

Physics and Astronomy Department “Galileo Galilei”, University of Padova, Italy

PREVIOUS POSITIONS

2021 – 2023 **Marie Curie Fellow. Project: THESYS.** GEPI, Observatoire de Paris, France

Maternity leave (Nov 2022 – March 2023)

2020 – 2021 **Researcher**, Lund Observatory, Lund, Sweden

Maternity leave (Feb 2020 – Oct 2020)

2015 – 2019 **Independent Research Fellow (MPIA prize fellowship)**, Galaxies and Cosmology Department, Max Planck Institute for Astronomy, Heidelberg, Germany

2012 – 2015 **Postdoctoral Fellow (Lady Davis Fellowship),**

Physics Department, Technion-Israel Institute of Technology, Haifa, Israel

EDUCATION

2012 **PhD in Astronomy.** Grade: Excellent

Department of Physics and Astrophysics, Sapienza University of Rome, Italy

Supervisor: Prof. Roberto Capuzzo-Dolcetta

2008 **Master's Degree in Astronomy and Astrophysics.** Grade: 110/110 summa cum laude

Department of Physics and Astrophysics, Sapienza University of Rome, Italy

FELLOWSHIPS AND GRANTS

2027 – **STARS@UniPD**, UniPD seeding grant for independent research (175,000EUR)

2026 – **Horizon 2020 MSCA Staff Exchange**, MEROPE, (100,000EUR as the scientific lead of the UniPD node)

2024 – **Ramon y Cajal Fellowship**, the most competitive Spanish national tenure track fellowship (350,000EUR, declined, ranked 13th over more than 350 applicants)

2021–2023 **Horizon 2020 MSCA Individual Fellowship** (200,000EUR, THESYS, grant agreement No 895174). Host institution: CNRS and GEPI, Observatoire de Paris, France.

2015 – 2019 **MPIA Prize Fellowship** (250,000EUR). Host institution: Galaxy and Cosmology Department, Max Planck Institute for Astronomy, Heidelberg, Germany.

2012 – 2014 **Lady Davis Fellowship** (18,000 USD). Host institution: Technion, Israel.

2012 **HPC-Europa2 grant** (3,000 EUR, 50,000 CPU hours and unlimited GPU hours for 3 months). Host institution: GEPI – Observatoire de Paris.

2008 – 2011 **PhD Scholarship** (75,000EUR). MIUR, Host institution: Sapienza Univ. of Rome.

PRIZES AND AWARDS

1) **“Immagine”** prize for the outstanding scientific career, awarded by “Associazione Immagine”, Latina, IT. Annual award for outstanding scientists (10/12/16).

- 2) “**Lievito 2016**” prize, awarded by “Associazione Lievito”, Latina, IT. Annual award for outstanding scientists (25/04/16).
- 3) “**Latina va in Scena**” prize for the outstanding scientific achievements. Award from Latina County in Italy (28/06/15).
- 4) **Formal recognition** (Italian: “Menzione d’Onore”) from the city of Pontinia (25/05/15).

SUCCESSFUL COMPUTING AND OBSERVING TIME PROPOSALS

- 2025 **Co-I** for **2 JWST** proposals and **4 HST** proposals (Pis: A. Milone, M.V. Legnardi, G. Cordoni, F. Muratore).
- 2021 **PI** of a SNIC computing proposal, granted resources 120,000 GPU/CPU hours on Swedish supercomputers. Project: “The origin of the Galactic nucleus”.
- 2020 **Co-I** for the **Hubble Space Telescope proposal Cycle 28** observing proposal “Digging into the mystery of the Galactic globular clusters M22 and NGC1851” (PI: Annalisa Calamida, STScI).
- 2017 **Co-PI** for a CINECA proposal, granted resources: 200000CPU/GPU hours (for 9 months). Project: “Formation of nuclear clusters through mergers of a realistic population of globular clusters”.
- 2014 **Co-PI** for a LinkSCEEM/Cy–Tera proposal, granted resources: 12000 GPU hours available for 12 months. Project: “Formation and evolution of dense stellar clusters”.
- 2014 **Co-I** for a GEMINI proposal, “The Milky Way nuclear star cluster as a benchmark for the structure and build-up of galactic nuclei” (PI: Anja Feldmeier-Krause, ESO).
- 2011 **Co-I** for a CFHT proposal, “The extended tidal tails of Palomar 14” (PI: Prof. D. Valls-Gabaud, Observatoire de Paris).

PUBLICATION SUMMARY: 10,600 citations (ADS, as of 08/2025)

- **h-index: 38**;
- **92** publications in peer-reviewed, high impact-factor journals:
- **12** as the **first author** including: **1 Nature** paper as the first author; **22** as the **second or third author (including 10 as the supervisor of PhD students)**.
- 2 white papers and 14 conference proceedings (10 refereed, 8 as the first author).

PUBLIC LECTURES, TALKS AND SEMINARS SUMMARY

- 10 invited talks at conferences (since 2014);
- 13 public outreach seminars/lectures (including a **TED^x talk**, since 2015).
- 22 invited colloquia/seminars at international institutes (since 2012),
- More than 50 press interviews, including some on gender equality in STEM.
- 32 contributed talks at conferences (since 2010).

INVITED CONFERENCE TALKS (10 talks)

- 2026 **Invited Talk**. “Traversing the Galactic Center in Space and Time”, Brno, Czech, 18-22/09. **Title:** “The Formation and Dynamical Evolution of the Galactic Nuclear Star Cluster”.
- 2023 **Invited Talk**. “A multi-wavelength view on globular clusters near and far: from JWST to the ELT”, Sexten, Italy, 3-7/07. **Title:** “Modelling the origin of multiple populations in globular clusters.”
- 2023 **Invited Talk**. “Phases Of Galactic Evolution As Traced By Stellar Populations And Star Clusters”, Sexten, Italy, 26-30/06. **Title:** “Modelling the origin of multiple populations in globular clusters.”
- 2022 **Invited review talk**. “The Puzzles of the Galactic Centre”, Heidelberg, Germany, 5-7/09. **Title:** “The Galactic nuclear star cluster and supermassive black hole formation and co-evolution: where do we stand?”. During the same conference, **I led a discussion section** as an expert on nuclear star cluster formation and evolution.
- 2019 **Invited talk**. “Uncovering the Physics of Formation of Globular Clusters and their Host Galaxies”, KITP, Santa Barbara, US, 11-14/05, postponed due to the Covid19 outbreak. **Title:** “The dynamics of merging globular clusters”.
- 2018 **Invited talk**. “Multiple populations in Globular clusters”, Sexten, Italy, 09-13/07. **Title:** “On the merger origin of metallicity spreads in globular clusters”.
- 2018 **Invited talk**. MPA Galaxy and Cosmology Retreat 2018, Lobbach, Germany, 14-

16/05. **Title:** “The chemo-dynamical evolution of the Solar System”.

2017 **Invited review talk.** “The amazing life of stars”, Cefalù, Italy, 04-08/09. **Title:** “The formation and evolution of Nuclear star clusters”.

2014 **Invited review talk.** “Growth and evolution of the Milky Way’s nuclear star cluster and its central black hole”, Alajar, Spain, 20-28/09. **Title:** “The formation and evolution of Nuclear star clusters”.

2014 **Invited talk.** “Stellar *N*-body Dynamics”, Sexten, Italy, 08-12/09. **Title:** “The formation and evolution of Nuclear star clusters”.

TEACHING EXPERIENCE

2025 – **Lecturer:** 1st year BSc, General Physics, Engineering Department, University of Padova, 88h.

2023 **Master’s internship lecturer:** Lecturer for a 1st year MSc internship project. Title of the project: “Dynamical evolution and fate of globular clusters in the Milky Way”. One student, 40h (2nd semester of the 2022-2023 Academic Year).

2022 **Lab insertion module lecturer:** “Dynamical evolution and fate of globular clusters in the Milky Way”, 1st year MSc. Two students, 40h (1st semester).

2021 **Lecturer,** 1st year MSc, “High-Energy Astrophysics”, Lund Observatory (Sweden). 15 students, 15h.

2020 **Lecturer,** 1st year MSc, “Stellar Structure and Evolution”, Lund Observatory (Sweden). 15 students, 3h;

2009 – 2010 **Teaching Assistant,** 3rd year BSc, *Numerical Methods in Astronomy*, Department of Physics and Astrophysics, Sapienza University of Rome, Italy. 50 students, 35h

SUPERVISION OF STUDENTS

8 PhD students co-supervised or mentored, 2 Master’s students and 1 Bachelor student supervised.

2022 – ongoing

(1) **Co-supervisor** (with S. Salvadori, UniFi): **PhD student Alice Mori.**

(2) **Co-supervisor** (with P. Di Matteo, GEPI): **PhD student Salvatore Ferrone.** Project: “The tidal evolution of Galactic globular clusters”. **One paper published:** Ferrone, S. Di Matteo, P., **Mastrobuono-Battisti, A.** et al., 2023, **A&A**, 673, A44.

(3) **Supervisor** (graduated: 04/10/2022, GEPI): **Master’s student Lisa Delit.** Project: “Globular cluster disruption and the link with the Galactic nucleus” (funded through my grant).

(4) **Co-supervisor** (with A. Milone, Padova Uni.; graduated 12/10/2022): **Master student Mariasole Maglione.** Project: “Multiple populations in globular clusters”, GEPI, Paris.

2021 – 2023

(1) **Co-supervisor** (with P. Di Matteo, GEPI): **PhD student Giulia Pagnini.** Project: “The early and late evolution of our Galaxy through the lens of globular clusters”. **One paper published:** Pagnini, G., Di Matteo, P., Khoperskov, S., **Mastrobuono-Battisti, A.** et al., 2023, **A&A**, 673, A86.

(2) **Co-supervisor** (with F. Calura, INAF-Bologna): **PhD student Elena Lacchin.** Project: “Mass loss in Magellanic Clouds Globular Clusters”. **One paper published:** Lacchin, E., Calura, F., Vesperini, E., **Mastrobuono-Battisti, A.**, 2022, **MNRAS**, 517, 1171.

2017 – 2021

Co-supervisor (with G. van de Ven and A. Sippel; graduated on 06/07/2021): **PhD student Francisco Aros.** Project: “Intermediate mass black holes in globular clusters”, MPIA, Germany. Graduated in July 2021 (magna cum laude). **Two published papers:**

1. Aros, F., Sippel, A., **Mastrobuono-Battisti, A.**, et al., 2020, **MNRAS**, 499, 4646

2. Aros, F., Sippel, A., **Mastrobuono-Battisti, A.**, et al., 2021, **MNRAS**, 508, 4385.

2017 – 2018: **Supervisor** (graduated on 19/06/2018): Bachelor student **Robin Herlan.** Project: “Dark matter density profiles in dwarf galaxies: the case of Sagittarius”, MPIA, Heidelberg, Germany. One paper published: Herlan, **Mastrobuono-Battisti**, Neumayer, **MNRAS**, **MNRAS**, 523, 2721.

2012 – 2017: Mentoring and co-supervision of **3 PhD students:**

- (1) **Federico Abbate** (Milan University, Italy, graduated in 2019). **One paper published:** Abbate, **Mastrobuono-Battisti** et al. 2018, MNRAS, 473, 927.
- (2) **Athanasia Tsatsi** (MPIA, Germany, graduated in 2016). **One paper published:** Tsatsi, **Mastrobuono-Battisti** et al. 2017, MNRAS, 464, 3720;
- (3) **Danor Aharon** (Technion, Israel, graduated in 2016). **One paper published:** Aharon, **Mastrobuono-Battisti** et al. 2016, ApJ, 823, 137.

ORGANISATION OF SCIENTIFIC MEETINGS

- 2023 **Co-Chair of the SOC and Co-I of the proposal** for the **EAS Special Session** “Understanding the connection between nuclear star clusters, black holes and galaxy evolution: merging observations and simulations”, **Kraków, Poland** (10-14/07/2023).
- 2021 **SOC member** for the conference “Multiple stellar populations in the next decade” (to be held in Padova, Italy, delayed due to the covid outbreak, <https://indico.ict.inaf.it/event/994/>)
- 2019 **SOC member and Co-I of the proposal** for the **EWASS special session** “The dynamics of stellar clusters”, Lyon (24-28/06/2019).
- 2018 **Principal organiser, chair of the SOC and PI of the funding proposal (15000 EUR granted)** for the conference “Survival of dense stellar clusters in the Milky Way system”, 85 participants, Heidelberg, Germany (19-23/11/2018).
- 2017 **(1) SOC member, discussion moderator and LOC co-chair** for the meeting: “Piercing the Galactic darkness”, 70 participants, Heidelberg, Germany (16-19/10/2017). **(2) Principal organiser, SOC and LOC member** for the MPIA Galaxy and Cosmology Department annual retreat, 90 participants, Schöntal, Germany (22-24/05/2017). **(3) SOC member and discussion moderator** for the meeting: “The exciting lives of galactic nuclei”, 40 participants, Ringberg, Germany (27/02-03/03/2017).

INSTITUTIONAL RESPONSIBILITIES

- 2023 – Member of the **“Lunar Gravitational Wave Antenna”** collaboration.
- 2022 – Member of the **ESO/ELT MOSAIC Science Team**.
- 2021 – **Co-organiser** of the “Paris Observatory Joint Galaxies and Cosmology Seminar”: <https://ggsapi.obspm.fr/seminars/>.
- 2021 **Referee and committee member** for the examination and evaluation of a **Master’s Thesis and a Bachelor’s Thesis** (Lund, Sweden). I refereed the Theses and I have been a panel member for the Thesis’ final evaluation.
- 2021 Member of the International Astronomical Union.
- 2018 – Member of **Gaia DPAC consortium (CU6 group)**, responsible for the **validation of the spectroscopic data in globular clusters**.
- 2016 – 2021 **Weekly group meetings organiser and chair** (Sweden and Germany).
- 2016 – Member of several **committees for the selection of future postdocs, PhD**

SELECTED PRESS COVERAGE (among more than 50 press articles, TV and radio interviews)

This is a partial list of the press releases concerning my work. They include interviews with journals, radio and television.

1. **List of citing articles (Nature):** <https://www.nature.com/articles/nature14333/metrics>
2. **National Geographic:** <https://tinyurl.com/3mwjyhy3>
3. **Discovery Channel (Seeker):** <https://tinyurl.com/2z42frf9>
4. **Science:** <http://www.sciencemag.org/news/2015/04/how-did-moon-really-form>
5. Ranked **#47 among the top 100 science stories of 2015** on Discovery magazine: <https://www.discovermagazine.com/the-sciences/the-moons-violent-birth>
6. **BBC:** <http://www.bbc.com/news/science-environment-32219494>
7. **NBC:** <https://tinyurl.com/2p8hdxth>
8. **Los Angeles Times:** <https://tinyurl.com/losAngTimes>
9. **Le Huffington Post:** <https://tinyurl.com/huffingtonpostfrance>
10. **Der Spiegel:** <https://tinyurl.com/derspiegelde>
11. **El Pais:** http://elpais.com/elpais/2015/04/08/ciencia/1428495111_666688.html

12. **Le Scienze** (Italian version of Scientific American): http://www.lescienze.it/news/2015/04/09/news/impatto_terra_formazione_luna_composizio ne-2559065/
13. **Il Corriere della Sera**: <https://tinyurl.com/ITcorrieredellasera>
14. **Technion press release**: <http://www.technion.ac.il/en/2015/04/the-origins-of-the-moon/>
15. **Swiss Television interview**: <https://tinyurl.com/Swiss-TV-Interview>
16. **Radio24, Giovani Talenti interview**: <https://tinyurl.com/yjsyd82f>
17. **GEPI press release on Palomar 5 tidal tails**: <https://tinyurl.com/gepitudaltails>
18. **Contribution to book chapters on outstanding women in science and technology**:
 - <https://tinyurl.com/Agnese-Interview>
 - <https://www.donnamoderna.com/news/speciali/donne-come-noi/donne-come-noi-il-libro>

REVIEWING ACTIVITIES

- 2022 –2023 Expert reviewer: **Hubble Space Telescope Cycle 30 and 31 TAC** (External Stellar Populations Panelist).
- 2013 – Referee for a number of **IOP Journals** as well as **MNRAS, A&A and A&AR**.

CAREER BREAKS AND IMPACT ON PRODUCTIVITY

- 2026 **Maternity leave** (July 2026 – December 2026, 5 months)
- 2022 **Maternity leave** (Nov 2022 – March 2023, 4 months)
- 2020 **Maternity leave** (Feb 2020 – Oct 2020, 8 months)

REFERENCES

- 1) **Dr. Paola Di Matteo**
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- 3) **Prof. Hagai Perets**
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- 4) **Dr. Ross P. Church**